TaxonVision-MLOps

Comprehensive Technical Implementation Guide

Systems Architecture, Infrastructure, and Codebase Reference

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October 11, 2026

Abstract

This technical report serves as an exhaustive, implementation-level architectural manual and code base reference for the TaxonVision-MLOps platform. Designed to document every structural component, configuration file, deployment manifest, mathematical loss formulation, and runtime daemon in the active repository, this document provides complete visibility into the physical server topology (`hive-mind` running Ubuntu 26.04.1 LTS, NVIDIA Driver 615.71.09, CUDA 13.4, and the NVIDIA GeForce RTX 5070 Ti with Blackwell SM 120), the lightweight Kubernetes (k3s) cluster partitioned into `jenkins-ci` and `taxon-vision` namespaces, the Cloudflare Zero Trust ingress tunnel, and the four-slot NVIDIA GPU time-slicing device plugin. Furthermore, this manual dissects the pure Pixi multi-environment toolchain, the automated DagsHub storage remote for Data Version Control (DVC) and MLflow tracking, the Cui et al. Class-Balanced Loss dynamics on biological long-tail Pareto distributions, the Optuna Bayesian hyperparameter optimisation engine, Split Conformal Prediction, Energy-Based Out-of-Distribution rejection, and the unified FastAPI and HTMX operator dashboard. Every configuration parameter, file extension, and container specification is strictly analysed based upon active codebase evidence.

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Chapter 1

Repository Architecture, Directory Hierarchy, and File Extension Semantics

1.1 Root Directory Topology

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1.3 Specialised File Formats and Storage Conventions

Chapter 2

Bare-Metal Server Environment and Pixi Toolchain Interlock

2.1 Physical Compute Node Baseline

The architectural foundation of the TaxonVision-MLOps ecosystem relies on a highly specified, bare-metal physical compute topology, designed to eliminate the hypervisor overhead typically associated with virtualised cloud environments. The principal workstation and training node, designated as `hive-mind`, operates on Ubuntu 26.04.1 LTS (codename 'resolute'), leveraging Linux Kernel 6.8+ to ensure advanced scheduler capabilities and extensive peripheral hardware support. The utilisation of an x86_64 architecture provides the necessary compatibility matrix for the compiled binary dependencies critical to the machine learning pipeline.

Central to the computational capacity of the `hive-mind` node is the integration of the NVIDIA GeForce RTX 5070 Ti hardware accelerator. This graphics processing unit represents the vanguard of the Blackwell architecture, characterised by Streaming Multiprocessor (SM) 120 and Compute Capability 12.0. Equipped with 16 Gigabytes of GDDR7 Video Random Access Memory (VRAM), the accelerator guarantees unprecedented memory bandwidth for high-resolution image batch processing. Operating within a 300-Watt Thermal Design Power (TDP) envelope, the RTX 5070 Ti communicates with the host processor via a Peripheral Component Interconnect Express (PCIe) 5.0 interface. This generation of the PCIe standard ensures a symmetrical bandwidth of 32 Giga-transfers per second (GT/s) per lane, effectively eliminating the data-starvation bottlenecks frequently observed during the asynchronous transfer of multi-gigabyte Parquet datasets from system memory to device VRAM.

In order to fully harness this computational capacity, the system necessitates a rigorous driver interlock. The host environment is strictly constrained to NVIDIA Driver version 615.71.09, synchronised with a Kernel Mode Driver (KMD) version 615.71.09 and a Compute Unified Device Architecture (CUDA) User Mode Driver (UMD) version 13.4. This configuration establishes a stable foundation for the CUDA 12.8 toolchain, which is non-negotiable for compiling custom kernels and executing PyTorch 2.7.0 operations optimally on the Blackwell microarchitecture.

In stark contrast to the high-performance `hive-mind` node, the ecosystem must also accommodate disparate developer workstation baselines. For instance, edge deployment simulations and initial algorithm prototyping may occur on legacy hardware, such as the NVIDIA Quadro T2000, which relies on the older Turing architecture (SM 75). The discrepancy in compute capabilities between the primary training node and auxiliary workstations mandates a flexible yet highly reproducible software environment encapsulation strategy, ensuring that algorithmic logic remains invariant irrespective of the underlying physical acceleration substrate.

The storage layer of the bare-metal host is engineered to saturate the PCIe 5.0 bus. The local

Non-Volatile Memory Express (NVMe) solid-state drive layout is partitioned deliberately to isolate operating system execution from the machine learning data plane. A dedicated high-speed partition is reserved exclusively for Parquet batch streaming and the caching of decoded image tensors. This partition ensures deterministic I/O bandwidth guarantees, preventing the operating system's background telemetry and logging operations from pre-empting the read operations of the data loader workers. The mathematical formulation of the I/O throughput must satisfy the condition where the data supply rate exceeds the GPU consumption rate, expressed as:

$$\mathcal{B}_{\text{read}} \geq \frac{N \times S_{\text{tensor}}}{\mathcal{T}_{\text{forward}} + \mathcal{T}_{\text{backward}}}$$

where $\mathcal{B}_{\text{read}}$ denotes the sustained bandwidth, N the batch size, S_{tensor} the physical size of the batch in memory, and the denominator represents the temporal duration of a single forward and backward propagation step. The strict hardware specification of the `hive-mind` node ensures this inequality is consistently maintained.

2.2 Multi-Target Toolchain and Pixi Environment Isolation

To eradicate the persistent ambiguities associated with traditional Python environment management—specifically the non-deterministic resolution algorithms of `pip` and the systemic fragility of `virtualenv`—the TaxonVision-MLOps architecture adopts the Pixi package manager. Built upon the Rattler resolver, Pixi provides pure binary Conda package management without the virtual environment overhead. The ecosystem enforces an absolute prohibition on the execution of bare `python`, `pip`, `poetry`, `uv`, and the creation of `.venv/` directories. All dependencies are hermetically sealed and resolved via SAT-based (Boolean satisfiability problem) dependency resolution mechanisms.

The project configuration, declared within `pyproject.toml`, establishes a dual-environment specification designed to cater seamlessly to both the hardware-accelerated local development context and the resource-constrained Continuous Integration (CI) runners. The `dev` environment is heavily augmented with GPU dependencies. It explicitly requests PyTorch 2.7.0, Torchvision 0.22.0, and Torchaudio 2.7.0, statically linked against the `cu128` (CUDA 12.8) distribution, which is imperative for compatibility with the SM 120 architecture of the RTX 5070 Ti. Additionally, the `dev` environment encompasses TensorRT and ONNX Runtime GPU dependencies to facilitate downstream model exportation and inference optimisation.

Conversely, the `ci-dev` environment is defined as a minimal, CPU-only footprint. It substitutes the heavy CUDA binaries with PyTorch 2.5.1 sourced from the CPU-only channel. This architectural bifurcation ensures that ephemeral CI runners, which lack physical GPU accelerators, can execute the test suite and validation scripts deterministically without incurring the bandwidth and storage penalties associated with downloading multi-gigabyte CUDA libraries.

The resolution logic prioritises specific package indices to prevent supply chain poisoning and ensure maximum performance. The channel priority resolution mandates that base system libraries and scientific utilities are sourced from `conda-forge`, whilst machine learning frameworks are pulled securely from the `pytorch` and `nvidia` channels. The exact configuration is formulated as a hierarchical fallback mechanism:

$$\mathcal{C} = \langle \texttt{conda-forge}, \texttt{pytorch}, \texttt{nvidia} \rangle$$

The cornerstone of this environmental reproducibility is the `pixi.lock` file. Unlike standard requirement files, `pixi.lock` serves as a cryptographic ledger. It records the exact uniform resource locators, the topological graph of the resolved dependencies, and the SHA-256 cryptographic hashes of every downloaded binary artefact. This immutability prevents upstream supply chain drift. When a developer or a deployment pipeline invokes `pixi install`, the system mathematically verifies the integrity of the fetched repodata against the locked hashes, throwing a fatal exception if a deviation is detected. This guarantees that the execution environment on the `hive-mind` node is bit-for-bit identical to the environment constructed inside a Docker container or an autonomous agent's sandbox.

2.3 Command Orchestration via Justfile

The standardisation of execution protocols is administered through `just`, a command runner that implements a declarative command interface. The repository's `Justfile` serves as the singular source of truth for automation recipes, functioning as an executable runbook that abstracts the complexities of the underlying Pixi environments.

A comprehensive analysis of the `Justfile` reveals a systematically categorised suite of recipes:

- `just check`: Triggers the execution of all configured pre-commit quality hooks across the entire repository. This recipe acts as the primary quality gate prior to version control integration.
- `just test`: Orchestrates the `pytest` framework, executing unit tests, integration tests, and critical mathematical invariant assertions, such as the conformal prediction marginal coverage tests.
- `just train`: Initiates the model training pipeline, specifically invoking the Command Line Interface (CLI) for the Class-Balanced Loss classifier. It dynamically maps parameters such as epoch count and feature extractor backbone (for example, `mobilenetv4_conv_small`).
- `just serve`: Boots the FastAPI asynchronous inference server and the corresponding Hypertext Markup Language (HTML) over HTMX operator dashboard, binding the process to localhost with live-reload capabilities.
- `just complexity`: Evaluates the source code using the `complexipy` utility, verifying that neither the cyclomatic complexity (the number of linearly independent paths through the source code) nor the cognitive complexity exceeds predefined thresholds.
- `just validate-okf`: Invokes the Open Knowledge Format (OKF v0.2) validation script to guarantee that all markdown documentation files adhere strictly to the project's metadata schema and link topology requirements.

The execution semantics of the `Justfile` enforce zero-wrapper execution. Every Python command is explicitly prefixed with `pixi run -frozen -e dev`. The `-frozen` flag enforces strict adherence to the `pixi.lock` file, preventing any inadvertent environment updates during execution. This explicit invocation avoids reliance on ambient environment variables or global state contamination.

Furthermore, the `Justfile` implements sophisticated environment variable inheritance and dry-run capabilities. Recipes securely interpolate secrets, such as DagsHub authentication tokens retrieved via the Linux Secret Service Application Programming Interface (API), directly into the execution context without exposing them in the terminal history. The declarative nature of the file, structured with logical dependencies (for example, `just setup` invoking private routines like `_setup-core`), ensures that complex multi-stage initialisations are executed idempotently.

2.4 Git Security and Pre-Commit Interception Engine

The preservation of codebase integrity and the attribution of modifications are strictly governed by a cryptographic commit interception engine. The architecture differentiates between two classes of actors: human operators (and local heuristic agents like Antigravity) and cloud-based autonomous agents (such as Jules).

For local workstation operations, the system mandates cryptographic commit signing invariants. Every commit must be accompanied by a valid GNU Privacy Guard (GPG) or Secure Shell (SSH) signature, executed via `git commit -S`. This cryptographic assertion guarantees non-repudiation, ensuring that every line of code can be mathematically traced back to a verified author. In contrast, the autonomous agent Jules operates within a constrained cloud sandbox where local signing keys cannot be securely provisioned. Therefore, Jules is instructed to bypass the local `-S` flag and push directly to a Pull Request feature branch. The commit is subsequently signed by the GitHub

Web-Flow entity upon the successful merge of the Pull Request, maintaining the chain of trust at the repository integration level.

Before any code modification is permitted to enter the Git object database, it must traverse a formidable gauntlet of static analysis checks defined within `.pre-commit-config.yaml`. This configuration specifies twenty discrete interception hooks, categorised functionally:

1. **Syntactic and Base Sanity Regulators:** Hooks such as `trailing-whitespace`, `end-of-file-fixer`, and `mixed-line-ending` enforce fundamental ASCII file hygiene. The `check-added-large-files` hook prevents the accidental inclusion of multi-megabyte binary assets, which must be routed through Data Version Control (DVC) instead.
2. **Cryptographic and Secret Auditing:** The `gitleaks` hook scans the staged diffs for high-entropy strings, API keys, and asymmetric private keys, preventing the inadvertent leakage of credentials into the version control history.
3. **Static Analysis and Type Enforcement:** The `ruff-check` and `ruff-format` hooks perform rapid static analysis and deterministic code formatting, replacing disparate tools like Flake8 and Black. Simultaneously, the `mypy` hook enforces strict gradual typing annotations across the `src/` and `scripts/` directories, ensuring that the Python interpreter's dynamic nature does not mask underlying type mismatches.
4. **Documentation and Typographical Invariants:** The `markdownlint` and `verify-markdown-rules` hooks ensure strict compliance with internal documentation protocols, notably Rule 04, which dictates absolute formatting standards for ASCII hyphens, unordered lists, and blank line padding. The `codespell` hook performs lexicon validation, explicitly instructed to ignore domain-specific nomenclature such as `taxa` and `darwincore`.
5. **Notebook Sanitisation:** The `nbstripout` hook intercepts Jupyter notebook commits, algorithmically stripping the ephemeral output cells and execution counts to ensure that only the semantic logic is tracked, thereby preventing massive Git repository bloat.
6. **Domain-Specific Compliance:** The `validate-okf` hook parses markdown frontmatter to ensure compliance with the Open Knowledge Format v0.2. The `audit-license-compliance` hook scans the data ingestion pipeline, verifying mathematically and legally that citizen-science observations harvested from external sources conform to the permissible open-source licenses.
7. **Toolchain Enforcement:** The bespoke `check-pure-pixi` hook runs a system-level bash assertion to verify the absence of `.venv` directories and `uv.lock` files. The `enforce-author-identity` hook inspects the local Git configuration to ensure that `commit.gpgsign` evaluates to true, aborting the transaction if the developer has attempted to bypass the signing mandate.

By interlocking the physical host topology with the deterministic Pixi resolver and the stringent pre-commit interception engine, the TaxonVision-MLOps ecosystem constructs an environment where computational reproducibility is not merely an aspiration, but a mathematically verified invariant.

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